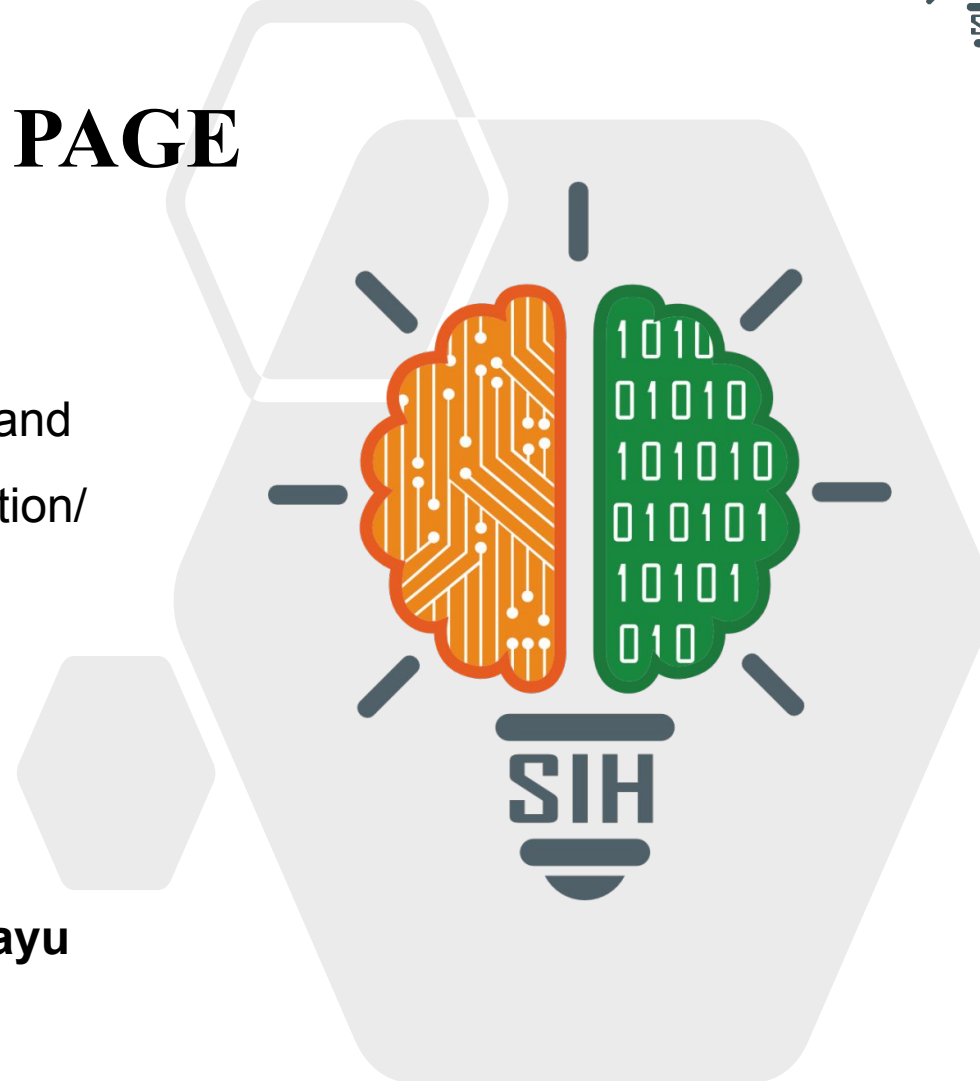


TITLE PAGE

- **Problem Statement ID – 1705**
- **Problem Statement Title-** Development and Optimization of AI model for Feature identification/Extraction from drone orthophotos.
- **Theme-** Robotics and Drones
- **PS Category-** Software
- **Team ID-** 13682
- **Team Name (Registered on portal) - Team Vaayu**



◆ Proposed Solution

1. Detailed Explanation of the Proposed Solution

- Develop a **cloud-based web portal** for uploading **drone orthophotos** from the **SVAMITVA Scheme**
- Integrate **Meta's Segment Anything Model (SAM)** for identifying key features like **building footprint extraction**, **road feature extraction**, and **water bodies extraction**.
- Use **YOLO, UNet and custom CNN models** for **image classification** of roofs (classify into **RCC, Tile, Tin** and other roof types)
- Calculation of built-up areas and abadi land for various services such as **solar energy calculation**, **property tax calculation**, etc
- Supports **exporting data** for **Geo-spatial Analysis** and provides output in desired formats like **.SHP (Shapefiles)** which can be used in GIS (Geographical Information Systems) software.

2. How It Addresses the Problem

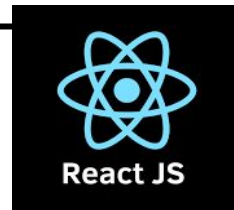
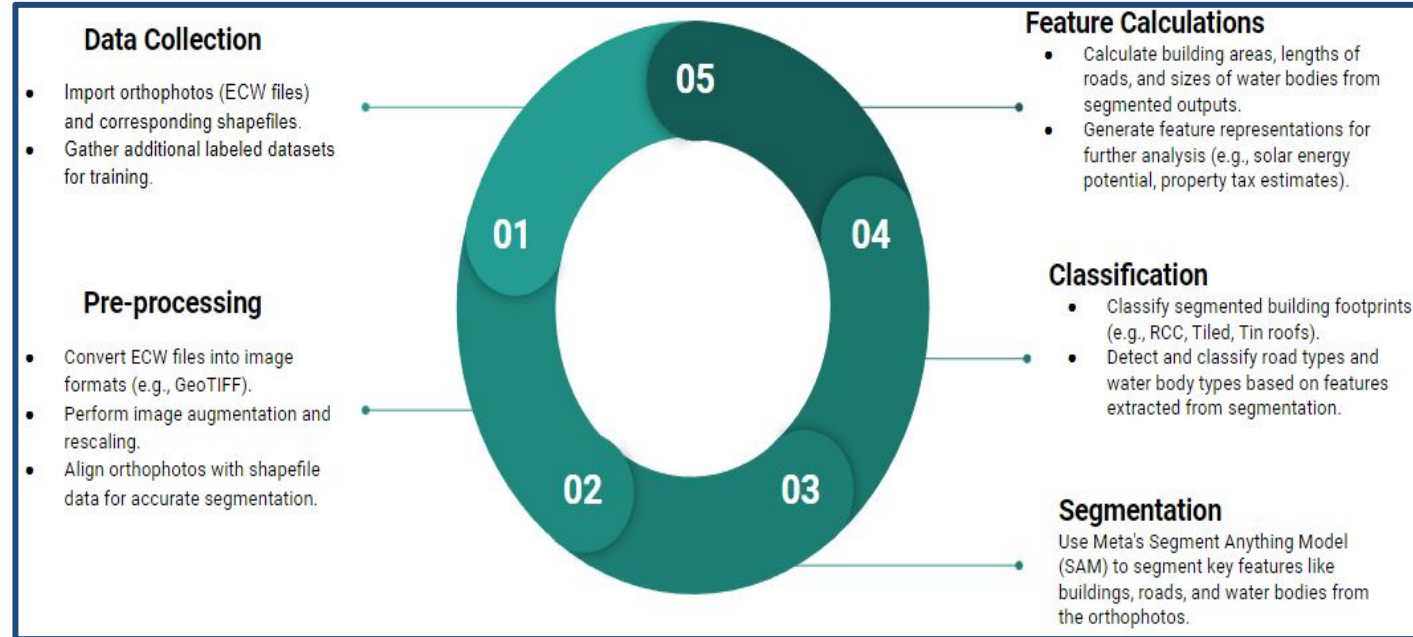
- Automates **high-precision feature extraction** from **drone orthophotos** using **computer vision techniques**.
- Reduces time, effort and resources in **detecting, segmenting and classifying** land features.
- Ensures accurate **image data** for better **land management** and **comprehensive village-level planning** using **Ground Control Points (GCPs)**
- Ensures extracted features are aligned with real-world geographical coordinates using **Geo-referencing integration**.

3. Innovation and Uniqueness of the Solution

- Leverages SAM's **state-of-the-art segmentation model** for adaptable **feature extraction** and **object detection**.
- Compatibility with file formats like **ECW (Enhanced Compression Wavelet)**, **TIFF (Tagged Image File Format)**, **GDB (GeoDataBase)** and **SHP (Shapefiles)**.
- Introduces **blockchain** for secure, authentic land ownership records.
- Uses **Augmented reality (AR)** for real-world visualization of orthophotos.
- Features a **chatbot** for interactive queries about segmented images using **LLMs and Vision Transformers**.



- **Frontend:** Next.js, Tailwind CSS, React, Three.js (for 3D visualization)
- **Backend:** Python, FastAPI, Solidity (for blockchain and smart contracts)
- **AI Models and other Libraries:** Meta's Segment Anything Model, Custom CNNs for Image Classification (ResNet, ImageNet), GPT/Gemini/Mistral API (for chatbot), GeoPandas, Fiona, pvlib, Pandas, Tensorflow, PyTorch, Numpy, Ultralytics and so on.
- **Database:** MongoDB, IPFS (for decentralized storage in blockchain), Vector Database for RAG (ChromaDB)
- **Cloud Infrastructure:** AWS EC2, Google Cloud VM, Amazon S3, Google Cloud Storage
- **Containerization:** Docker
- **Augmented Reality:** Unity/Unreal Engine (for AR visualizations)
- **Blockchain:** Ethereum, Solidity, IPFS (for secure, decentralized land ownership records)



Technical Feasibility

- ❑ **Strengths:** Drone imagery from **SVAMITVA**; **AI/ML techniques** (CNNs, Image Segmentation); use of **Segment Anything Model** for **feature extraction**.
- ❑ **Challenges:** Ensuring data quality; high computational requirements for deep learning models and a **target accuracy** of 95%.
- ❑ **Overcoming Challenges:** Use **fine-tuning**, **transfer learning**, **cloud computing**, and **generative AI** to optimize training and deployment; implement **chatbot**.

Operational Feasibility

- ❑ **Strengths:** Integration with **Geospatial Information Systems (GIS)** platforms; cloud-based deployment for scalability; **blockchain** for secure land ownership records; **Augmented Reality** for visualization.
- ❑ **Challenges:** Custom development needed for seamless integration; ongoing maintenance required; ensuring interoperability with blockchain.
- ❑ **Overcoming Challenges:** Implement a robust maintenance plan

Economic Feasibility

- ❑ **Strengths:** **Cost-effective development** using open-source tools; long-term ROI through automation, **efficiency gains**.
- ❑ **Challenges:** High initial investment and potential ongoing cloud and GPU (**Graphics Processing Unit**) costs.
- ❑ **Overcoming Challenges:** Leverage **pre-trained models**, optimize cloud usage, and implement **smart contracts** for efficient resource management.

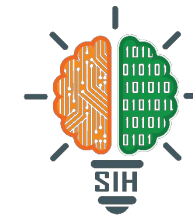
Legal and Ethical Feasibility

- ❑ **Strengths:** Manageable **data privacy** concerns with appropriate **anonymization**; legal viability; **blockchain** ensures data authenticity.
- ❑ **Challenges:** Potential **legal concerns** from communities; risk of biases in AI model; **ensuring ethical use** of blockchain data.
- ❑ **Overcoming Challenges:** Clear communication, legal frameworks, rigorous testing for fairness; ensure transparency in AI, blockchain, and AR applications.

Social and Environmental Viability

- ❑ **Strengths:** Significant community benefits; **minimal environmental impact**; enhanced **data security**; AR adds a new dimension to community engagement.
- ❑ **Challenges:** Ensuring public trust, **transparency**
- ❑ **Overcoming Challenges:** Engage **stakeholders**, maintain transparency, and **educate communities** on the benefits of **AI**, **blockchain**, and **AR**.

IMPACT AND BENEFITS



Impact on End-users

- **Record Management:** Provides rural communities with formalized property ownership and access to credit through blockchain-based records, enhancing security and authenticity.
- **Dispute Resolution:** Reduces property-related disputes with precise land records and secure ownership information.
- **Efficiency:** Streamlines land surveying processes and feature extraction using **CNN** and **AI**, saving time and resources for local administrations.
- **Interactive Assistance:** Enhances user experience with a chatbot on the interactive web portal, enabling users to query features in images (e.g. : “How many lakes are in this village?”).

Benefits of the Solution

- **Social:** Improves social equity by formalizing land ownership and reducing conflicts; chatbot interaction provides users with easy access to information about their land.
- **Economic:** Boosts economic growth through accurate property tax collection and enhanced access to financial services.
- **Administration:** Supports sustainable village level planning by providing accurate, detailed orthophotos and visualization..
- **Generative AI & AR:** AR allows for the visualization of orthophotos in real-world settings, offering unique insights and enhanced spatial understanding of land features.

- ❑ [Kirillov, Alexander, et al. "Segment anything." *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 2023.](#)
- ❑ [Dornaika, Fadi, et al. "Building detection from orthophotos using a machine learning approach: An empirical study on image segmentation and descriptors." *Expert Systems with Applications* 58 \(2016\): 130-142.](#)
- ❑ [Sameen, Maher Ibrahim, and Biswajeet Pradhan. "A novel road segmentation technique from orthophotos using deep convolutional autoencoders." *Korean Journal of Remote Sensing* 33.4 \(2017\): 423-436.](#)
- ❑ [Kudelka, Milos, et al. "Orthophoto feature extraction and clustering." *Neural Network World* 22.2 \(2012\): 103.](#)
- ❑ [Yiğit, Abdurahman Yasin, and Murat Uysal. "Automatic road detection from orthophoto images." *Mersin Photogrammetry Journal* 2.1 \(2020\): 10-17.](#)
- ❑ [Dornaika, Fadi, et al. "A comparative study of image segmentation algorithms and descriptors for building detection." *Handbook of neural computation*. Academic Press, 2017. 591-606.](#)
- ❑ [Abdollahi, Abolfazl, and Biswajeet Pradhan. "Integrated technique of segmentation and classification methods with connected components analysis for road extraction from orthophoto images." *Expert Systems with Applications* 176 \(2021\): 114908.](#)